

Nonlinear finite element analysis of elastic water storage tanks



Kalyan Kumar Mandal*, Damodar Maity

Department of Civil Engineering, Indian Institute of Technology Kharagpur, India

ARTICLE INFO

Article history:

Received 19 December 2014

Revised 28 April 2015

Accepted 30 April 2015

Available online 6 June 2015

Keywords:

Finite element method

Fluid structure interaction

Nonlinear hydrodynamic pressure

Navier–Stokes equation

Earthquake excitation

ABSTRACT

The present paper deals with the nonlinear finite element analysis of elastic water tanks. Both fluid and tank walls are discretized and modeled by two dimensional eight-node isoperimetric elements. In the governing equations, displacement for tank walls and pressure for the fluid domain are considered as independent nodal variables. The nonlinear term for the convective acceleration in Navier–Stokes equations is incorporated in the present analysis. The nonlinearity effects of the fluid are studied considering excitations both harmonic of various frequencies and random. The hydrodynamic pressure on tank wall is presented both from linear and nonlinear analysis for a comparison. The results show that the convective nonlinearity increases the magnitude of hydrodynamic pressure to a considerable amount when the excitation frequency approaches to the fundamental frequency of the water tank. The magnitude of hydrodynamic pressure in nonlinear analysis is quite large compare to that in linear analysis when the distance between the two vertical walls are relatively closer. However, the increase in pressure is insignificant when the tank is vibrated with a frequency higher than the frequency of the fluid domain. The seismic analysis results show that the distribution of nonlinear hydrodynamic pressure is almost similar to the linear pressure due to ground excitation.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Tanks are commonly used to store water and various fluids in the oil industry. Damages in tanks may cause a loss of liquid content, which could result in economic damage, as well as in long-term contamination of soil for tanks resting on soil. For the design of an earthquake-resistant water tank, it is important to use a rational and reliable dynamic analysis procedure. The analysis procedure should be capable of evaluating the dynamic deformations and hydrodynamic pressure on tank wall under probable ground motion.

For simplification of the analytical procedures, the tank walls are generally considered to be rigid. Housner [1] presented analytical solutions for rectangular and cylindrical rigid tanks under horizontal ground excitations. An analytical or a closed-form solution cannot account for the arbitrary geometry of the tank. This problem can be efficiently tackled with finite element technique. For finite element modeling of fluid, different variables such as displacement, pressure, velocity potential etc. are considered. Fenves and Vargas [2] considered velocity and pressure as nodal variables to represent the governing equations of fluid. Zienkiewicz et al. [3] represented the fluid domain in terms of a

displacement potential. Pelecanoset et al. [4] considered displacement as nodal variable to model the reservoir. Bouaanani and Lu [5] considered the velocity potential be the nodal variable for the analysis of reservoir. Maity and Bhattacharyya [6] and Gogoi and Maity [7] considered pressure to be nodal degree of freedom in the finite element modeling of the fluid.

Tung [8] studied the hydrodynamic pressure distribution on a rigid, submerged, cylindrical storage tank subjected to horizontal base excitation. The author considered fluid to be incompressible. Williams and Moubayed [9] calculated hydrodynamic pressure on rigid tank walls by boundary integral method due to horizontal and vertical excitations considering fluid to be linearly compressible. The effects of a bottom-mounted rectangular block on the sloshing characteristics of the fluid in rectangular tanks were investigated by Choun and Yun [10] using the linear water wave theory. The hydrodynamic pressure exerted on the tank wall and on the mounted block is changed significantly as the block becomes large and moves toward the wall. Consequent experimental study by Panigrahy et al. [11] showed that the pressure exerted on the walls varies in a similar nature as that of the applied excitation and the pressure at the wall showed a considerable fluctuation near the free surface of the liquid compared to the deeper surfaces in the tank. A time-independent finite difference method was developed by Hua et al. [12] to simulate fluid sloshing and sloshing induced hydrodynamic pressure on tank walls. The

* Corresponding author.

E-mail address: kkma_iitkgp@yahoo.co.in (K.K. Mandal).

authors showed that the horizontal hydrodynamic force due to sloshing waves acting on the mid-section of the wall of the tank is dominated by the added mass effect when the excitation frequency is larger than four times the lowest natural frequency of the tank. They also concluded that sloshing waves plays a key effect on the horizontal sloshing-induced force if the excitation frequency is less than four times the fundamental frequency of the tank. However, all the above numerical simulations are carried out considering tank walls to be rigid. The deformations, stresses and hydrodynamic pressure will be more realistic if the elastic properties of tank walls and wall–fluid interaction are considered appropriately.

Haroun and Tayel [13] analyzed the earthquake response of cylindrical liquid storage tanks under vertical excitations and showed that structures are more susceptible to the increase in hoop for vertical excitation. Similar study on cylindrical liquid storage tank was carried out by Veletsos and Tang [14] and Haroun and Abou [15] considering tank walls to be flexible. Kianoush and Chen [16] used finite element technique to obtain hydrodynamic pressure on a rectangular tank due to vertical excitation. Park et al. [17] used the coupled boundary-finite element method to study the response of similar rectangular tanks. Chen and Kianoush [18] calculated hydrodynamic pressure on rectangular tank wall by lumped mass approach as well as consistent mass approach and their results showed that the consistent mass approach reduces the response of liquid containing structures as compared to that of lumped mass approach. Moslemi and Kianoush [19] studied the influences of sloshing of free surface, tank wall flexibility, vertical ground acceleration, tank aspect ratio and base fixity on hydrodynamic pressure. The authors also compared the results with those calculated from well-proved analytical methods and concluded that current design procedure in estimating the hydrodynamic pressure was too conservative. The estimated hydrodynamic pressure will be more realistic particularly for strong ground motion when the fluid nonlinearity is considered properly.

A nonlinear liquid sloshing inside a partially filled rectangular tank was investigated by Celebi and Akyildiz [20]. Authors used volume of fluid technique to track the free surface and the model solved the complete Navier–Stokes equations in primitive variables by use of the finite difference approximations. Similar nonlinear sloshing for cylindrical water tank was determined by Barrios et al. [21]. Virella et al. [22] used linear and nonlinear wave theories to determine the linear and nonlinear hydrodynamic pressure on rigid tank walls using the finite element software ABAQUS. The pressure and displacement were considered as unknown variables for linear and nonlinear model respectively. A plane strain formulation with large displacements was used for nonlinear modeling and their study showed that the nonlinearity does not have significant effects on the natural sloshing periods. The linear wave theory conservatively estimated the magnitude of the pressure distribution, whereas larger resultant pressure is obtained if nonlinear theory is considered.

The exhaustive literature review reveals that no study has been carried out on the nonlinear behavior of water storage tanks considering tank flexibility. The accurate behavior of the water tank can be achieved if the coupled fluid–structure interaction effects are taken into consideration in the analysis. Generally, the displacement based finite element method is used to model the tank–water coupled system. However, many researchers solved Navier–Stokes equation with pressure term as variable parameter only in their modeling. In the present study, the nonlinear convective terms in the Navier–Stokes equation is taken into account in an iterative manner while solving the fluid equation. Water in the tank is assumed to be compressible. A pressure based finite element model is developed to obtain nonlinear response of fluid domain. The tank wall is modeled in two dimensions and coupled

with the fluid equation. A computer code in MATLAB environment has been developed to obtain hydrodynamic pressure, displacement and stresses on tank wall. The study is carried out for different amplitudes and frequencies of harmonic excitation and earthquake excitation to investigate the degree of nonlinear effects. The effects of fluid nonlinearity are also studied for different lengths of the tank.

2. Theoretical formulations for fluid

The state of stress for a Newtonian fluid is defined by an isotropic tensor as [23]

$$T_{ij} = -p\delta_{ij} + T'_{ij} \quad (1)$$

where T_{ij} is total stress, T'_{ij} is viscous stress tensor which depends only on the rate of deformation in such a way that the value becomes zero when the fluid is under rigid body motion or rest. The variable p is defined as hydrodynamic pressure whose value is independent explicitly on the rate of deformation and δ_{ij} is Kronecker delta. For isotropic linear elastic material, the most general form of T'_{ij} is

$$T'_{ij} = \lambda\Delta\delta_{ij} + 2\mu D_{ij} \quad (2)$$

Where μ and λ are two material constants. μ is known as first coefficient of viscosity or viscosity and $(\lambda + 2\mu/3)$ is second coefficient of viscosity or bulk viscosity. D_{ij} is the rate of deformation tensor and is expressed as

$$D_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial y_j} + \frac{\partial v_j}{\partial x_i} \right) \quad \text{and} \quad \Delta = D_{11} + D_{22} + D_{33} \quad (3)$$

Thus, the total stress tensor becomes

$$T_{ij} = -p\delta_{ij} + \lambda\Delta\delta_{ij} + 2\mu D_{ij} \quad (4)$$

For compressible fluid, bulk viscosity $(\lambda + 2\mu/3)$ is zero. Thus, Eq. (4) becomes

$$T_{ij} = -p\delta_{ij} - \frac{2\mu}{3}\Delta\delta_{ij} + 2\mu D_{ij} \quad (5)$$

If the viscosity of fluid is neglected, Eq. (5) becomes

$$T_{ij} = -p\delta_{ij} \quad (6)$$

Generalized Navier–Stokes equations of motion are given by

$$\rho \left(\frac{\partial v_i}{\partial t} + v_j \frac{\partial v_i}{\partial x_j} \right) = \frac{\partial T_{ij}}{\partial x_j} + \rho B_i \quad (7)$$

where B_i is the body force and ρ is the mass density of fluid. Substituting Eq. (6) in Eq. (7) the following relations are obtained.

$$\rho \left(\frac{\partial v_i}{\partial t} + v_j \frac{\partial v_i}{\partial x_j} \right) = \rho B_i - \frac{\partial p}{\partial x_i} \quad (8)$$

If u and v are the velocity components along x and y axes respectively and f_x and f_y are body forces along x and y direction respectively, the equation of motion may be written as

$$\frac{1}{\rho} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = f_x \quad (9)$$

$$\frac{1}{\rho} \frac{\partial p}{\partial y} + \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = f_y \quad (10)$$

Neglecting the body forces, Eqs. (9) and (10) become

$$\frac{1}{\rho} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = 0 \quad (11)$$

$$\frac{1}{\rho} \frac{\partial p}{\partial y} + \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = 0 \quad (12)$$

The continuity equation of fluid in two dimensions is expressed as [24]:

$$\frac{\partial p}{\partial t} + \rho c^2 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0 \quad (13)$$

where c is the acoustic wave speed in fluid. Now, differentiating Eqs. (11) and (12) with respect to x and y respectively, the following relations are obtained.

$$\frac{1}{\rho} \frac{\partial^2 p}{\partial x^2} + \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right) + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + u \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + v \frac{\partial^2 u}{\partial x \partial y} = 0 \quad (14)$$

$$\frac{1}{\rho} \frac{\partial^2 p}{\partial y^2} + \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial t} \right) + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} + v \frac{\partial^2 v}{\partial y^2} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} + u \frac{\partial^2 v}{\partial x \partial y} = 0 \quad (15)$$

Adding Eqs. (14) and (15) the following expression is finally arrived.

$$\begin{aligned} \frac{1}{\rho} \frac{\partial^2 p}{\partial x^2} + \frac{1}{\rho} \frac{\partial^2 p}{\partial y^2} + \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial t} \right) + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} \\ + u \frac{\partial^2 u}{\partial x^2} + v \frac{\partial^2 v}{\partial y^2} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} + v \frac{\partial^2 u}{\partial x \partial y} + u \frac{\partial^2 v}{\partial x \partial y} = 0 \end{aligned} \quad (16)$$

Differentiating Eq. (13) with respect to time, the following expression can be obtained.

$$\frac{\partial^2 p}{\partial t^2} + \rho c^2 \left\{ \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial t} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial t} \right) \right\} = 0 \quad (17)$$

Thus, from Eqs. (16) and (17), one can find the following expression:

$$\begin{aligned} \frac{1}{\rho} \frac{\partial^2 p}{\partial x^2} + \frac{1}{\rho} \frac{\partial^2 p}{\partial y^2} - \frac{1}{\rho c^2} \left(\frac{\partial^2 p}{\partial t^2} \right) + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} + u \frac{\partial^2 u}{\partial x^2} + v \frac{\partial^2 v}{\partial y^2} \\ + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} + v \frac{\partial^2 u}{\partial x \partial y} + u \frac{\partial^2 v}{\partial x \partial y} = 0 \end{aligned} \quad (18)$$

Neglecting higher order terms in Eq. (18), the equation becomes

$$\frac{1}{\rho} \frac{\partial^2 p}{\partial x^2} + \frac{1}{\rho} \frac{\partial^2 p}{\partial y^2} - \frac{1}{\rho c^2} \left(\frac{\partial^2 p}{\partial t^2} \right) + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} = 0 \quad (19)$$

If the nonlinear convective terms are neglected Eq. (19) further becomes

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = \frac{1}{c^2} \left(\frac{\partial^2 p}{\partial t^2} \right) \quad (20)$$

The pressure distribution in the fluid domain may be obtained by solving Eq. (19) with the following boundary conditions. A typical geometry of tank-water system is shown in Fig. 1.

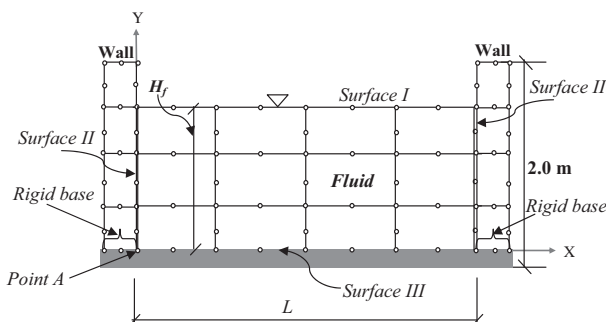


Fig. 1. A typical finite element discretization of tank-water system.

(i) At surface I

Considering the effect of surface wave of the fluid, the boundary condition of the free surface is taken as

$$\frac{1}{g} \ddot{p} + \frac{\partial p}{\partial y} = 0 \quad (21)$$

(ii) At surface II

At water-tank wall interface, the pressure should satisfy

$$\frac{\partial p}{\partial n}(0, y, t) = \rho_f a e^{i\omega t} \quad (22)$$

where $a e^{i\omega t}$ is the horizontal component of the ground acceleration in which, ω is the circular frequency of vibration and $i = \sqrt{-1}$, n is the outwardly directed normal to the element surface along the interface. ρ_f is the mass density of the fluid.

(iii) At surface III

This surface is considered as rigid surface and thus pressure should satisfy the following condition

$$\frac{\partial p}{\partial n}(x, 0, t) = 0.0 \quad (23)$$

2.1. Finite element formulation for fluid domain

By Galerkin approach and assuming pressure to be the nodal unknown variable, the discretised form of Eq. (19) may be written as

$$\begin{aligned} \int_{\Omega} N \left[\frac{1}{\rho} \frac{\partial^2 p}{\partial x^2} + \frac{1}{\rho} \frac{\partial^2 p}{\partial y^2} - \frac{1}{\rho c^2} \left(\frac{\partial^2 p}{\partial t^2} \right) \right. \\ \left. + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} \right] d\Omega = 0 \end{aligned} \quad (24)$$

where N is the interpolation function for the fluid and Ω is the region under consideration. Using Green's theorem Eq. (24) may be transformed to

$$\begin{aligned} \frac{1}{\rho} \int_{\Gamma} N \left(\frac{\partial p}{\partial x} + \frac{\partial p}{\partial y} \right) d\Gamma - \frac{1}{\rho} \int_{\Omega} \left(\frac{\partial N}{\partial x} \frac{\partial p}{\partial x} + \frac{\partial N}{\partial y} \frac{\partial p}{\partial y} \right) d\Omega \\ - \frac{1}{\rho c^2} \int_{\Omega} N \frac{\partial^2 p}{\partial t^2} d\Omega + \int_{\Omega} N \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} \right) d\Omega \\ + \int_{\Omega} N \left(\frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} \right) d\Omega = 0 \end{aligned} \quad (25)$$

Rearranging Eq. (25), the following expression may obtained

$$\begin{aligned} \frac{1}{\rho} \int_{\Omega} \left(\frac{\partial N}{\partial x} \frac{\partial p}{\partial x} + \frac{\partial N}{\partial y} \frac{\partial p}{\partial y} \right) d\Omega + \frac{1}{\rho c^2} \int_{\Omega} N \frac{\partial^2 p}{\partial t^2} d\Omega \\ - \int_{\Omega} N \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} \right) d\Omega - \int_{\Omega} N \left(\frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} \right) d\Omega \\ = \frac{1}{\rho} \int_{\Gamma} N \left(\frac{\partial p}{\partial x} + \frac{\partial p}{\partial y} \right) d\Gamma \end{aligned} \quad (26)$$

Now,

$$\frac{1}{\rho} \int_{\Omega} \left(\frac{\partial N}{\partial x} \frac{\partial p}{\partial x} + \frac{\partial N}{\partial y} \frac{\partial p}{\partial y} \right) d\Omega = \frac{1}{\rho} \int_{\Omega} \left[\frac{\partial N}{\partial x} \frac{\partial N}{\partial y} \right] \left[\frac{\partial p}{\partial x} \frac{\partial p}{\partial y} \right] d\Omega = \frac{1}{\rho} \int_{\Omega} B^T B \bar{p} d\Omega \quad (27)$$

where

$$\begin{bmatrix} \frac{\partial p}{\partial x} \\ \frac{\partial p}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial N}{\partial x} \\ \frac{\partial N}{\partial y} \end{bmatrix} \bar{p} \quad (28)$$

Here, $\bar{p}$ is the nodal pressure. The second term in Eq. (26) becomes

$$\frac{1}{\rho c^2} \int_{\Omega} N \frac{\partial^2 p}{\partial t^2} d\Omega = \frac{1}{\rho c^2} \int_{\Omega} N^T N \ddot{p} d\Omega \quad (29)$$

The third term in Eq. (26) becomes

$$\int_{\Omega} N \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} \right) d\Omega = \int_{\Omega} N^T \left[\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} \right] \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \end{bmatrix} d\Omega \quad (30)$$

Again,

$$\begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} = J^{-1} \begin{bmatrix} \frac{\partial u}{\partial \xi} \\ \frac{\partial u}{\partial \eta} \end{bmatrix} \quad (31)$$

where

$$J = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} \end{bmatrix} \quad \text{and} \quad J^{-1} = \begin{bmatrix} J_{11}^* & J_{12}^* \\ J_{21}^* & J_{22}^* \end{bmatrix} \quad (32)$$

$$\frac{\partial u}{\partial x} = J_{11}^* \frac{\partial u}{\partial \xi} + J_{12}^* \frac{\partial u}{\partial \eta} \quad (33)$$

$$\frac{\partial u}{\partial y} = J_{21}^* \frac{\partial u}{\partial \xi} + J_{22}^* \frac{\partial u}{\partial \eta} \quad (34)$$

Similarly,

$$\frac{\partial v}{\partial x} = J_{11}^* \frac{\partial v}{\partial \xi} + J_{12}^* \frac{\partial v}{\partial \eta} \quad (35)$$

$$\frac{\partial v}{\partial y} = J_{21}^* \frac{\partial v}{\partial \xi} + J_{22}^* \frac{\partial v}{\partial \eta} \quad (36)$$

Now,

$$\begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \end{bmatrix} = \begin{bmatrix} J_{11}^* & J_{12}^* & 0 & 0 \\ 0 & 0 & J_{21}^* & J_{22}^* \end{bmatrix} \begin{bmatrix} \frac{\partial u}{\partial \xi} \\ \frac{\partial u}{\partial \eta} \\ \frac{\partial v}{\partial \xi} \\ \frac{\partial v}{\partial \eta} \end{bmatrix} = C_1 \begin{Bmatrix} u \\ v \end{Bmatrix} \quad (37)$$

where

$$C_1 = \begin{bmatrix} J_{11}^* & J_{12}^* & 0 & 0 \\ 0 & 0 & J_{21}^* & J_{22}^* \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix} \quad (38)$$

where

$$A = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \dots & \frac{\partial N_n}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \dots & \frac{\partial N_n}{\partial \eta} \end{bmatrix} \quad (39)$$

Substituting the above in Eq. (30), the following expression is arrived.

$$\int_{\Omega} N \left(\frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} \right) d\Omega = \int_{\Omega} N^T [d_1^T C_1^T C_1 d_1] d\Omega \quad (40)$$

where

$$d_1 = \begin{Bmatrix} u \\ v \end{Bmatrix} \quad (41)$$

$$\int_{\Omega} N \left(\frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right) d\Omega = \int_{\Omega} N^T \left[\frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right] \begin{bmatrix} \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} d\Omega \quad (42)$$

$$\begin{bmatrix} \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} \end{bmatrix} = \begin{bmatrix} J_{21}^* & J_{22}^* & 0 & 0 \\ 0 & 0 & J_{11}^* & J_{12}^* \end{bmatrix} \begin{bmatrix} \frac{\partial u}{\partial \xi} \\ \frac{\partial u}{\partial \eta} \\ \frac{\partial v}{\partial \xi} \\ \frac{\partial v}{\partial \eta} \end{bmatrix} = C_2 \begin{Bmatrix} u \\ v \end{Bmatrix} \quad (43)$$

where

$$C_2 = \begin{bmatrix} J_{21}^* & J_{22}^* & 0 & 0 \\ 0 & 0 & J_{11}^* & J_{12}^* \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix} \quad (44)$$

and

$$\begin{bmatrix} \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} = \begin{bmatrix} J_{11}^* & J_{12}^* & 0 & 0 \\ 0 & 0 & J_{21}^* & J_{22}^* \end{bmatrix} \begin{bmatrix} \frac{\partial v}{\partial \xi} \\ \frac{\partial v}{\partial \eta} \\ \frac{\partial u}{\partial \xi} \\ \frac{\partial u}{\partial \eta} \end{bmatrix} = C_1 \begin{Bmatrix} v \\ u \end{Bmatrix} \quad (45)$$

$$\int_{\Omega} N \left(\frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right) d\Omega = \int_{\Omega} N^T [d_1^T C_2^T C_1 d_2] d\Omega \quad (46)$$

Here,

$$d_2 = \begin{Bmatrix} v \\ u \end{Bmatrix} \quad (47)$$

and

$$\frac{1}{\rho} \int_{\Gamma} N \left(\frac{\partial p}{\partial x} + \frac{\partial p}{\partial x} \right) d\Gamma = \frac{1}{\rho} \int_{\Gamma} N^T \frac{\partial p}{\partial n} d\Gamma \quad (48)$$

The whole system Eq. (26) may be written in matrix form as

$$[\bar{E}] \{\dot{P}\} + [\bar{G}] \{P\} = \{F_I\} - \{F_{nl}\} \quad (49)$$

If the nonlinear convective terms are neglected Eq. (49) becomes

$$[\bar{E}] \{\dot{P}\} + [\bar{G}] \{P\} = \{F_I\} \quad (50)$$

The matrices in Eq. (49) may be expressed as follows

$$[\bar{E}] = \frac{1}{c^2} \int_{\Omega} [N^T] [N] d\Omega \quad (51)$$

$$[\bar{G}] = \int_{\Omega} [B^T] [B] d\Omega \quad (52)$$

$$\{F_{nl}\} = \int_{\Omega} N^T [d_1^T C_1^T C_1 d_1] d\Omega + \int_{\Omega} N^T [d_1^T C_2^T C_1 d_2] d\Omega \quad (53)$$

$$\{F_I\} = \int_{\Gamma} [N^T] \frac{\partial p}{\partial n} d\Gamma = \{F_f\} + \{F_{fs}\} + \{F_{fb}\} \quad (54)$$

Here the subscript f , fs and fb stand for the free surface, fluid–wall interface and fluid–bed interface respectively. For surface wave, Eq. (21) may be written in finite element form as

$$\{F_f\} = -\frac{1}{g} [R_f] \{\ddot{p}\} \quad (55)$$

In which,

$$[R_f] = \sum_{\Gamma_f} [N_r]^T [N_r] d\Gamma \quad (56)$$

At the fluid–structure interface, if $\{a\}$ is the vector of nodal accelerations of generalized coordinates, $\{F_{fs}\}$ may be expressed as

$$\{F_{fs}\} = -\rho [R_{fs}] \{a\} \quad (57)$$

In which,

$$[R_{fs}] = \sum_{\Gamma_{fs}} [N_r]^T [T] [N_r] d\Gamma \quad (58)$$

Here, $[T]$ is the transformation matrix at fluid structure interface and N_r is the shape function of fluid. At fluid–bed interface

$$\{F_{fb}\} = 0 \quad (59)$$

After substitution all terms, Eq. (49) becomes

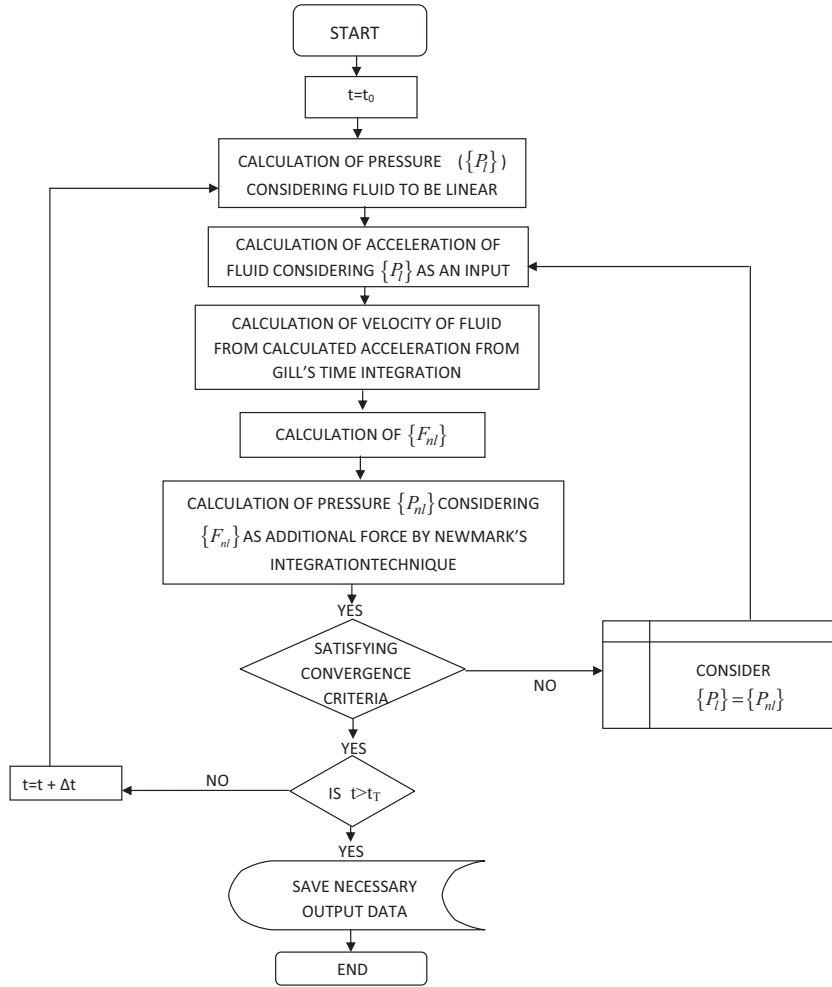


Fig. 2. Flow chart for nonlinear analysis of hydrodynamic pressure.

$$[E]\{\ddot{P}\} + [A]\{\dot{P}\} + [G]\{P\} = \{F_l\} - \{F_{nl}\} \quad (60)$$

where

$$[E] = [\bar{E}] + \frac{1}{g}[R_f] \quad (61)$$

$$\{F_l\} = -\rho[R_{fs}]\{a\} \quad (62)$$

$$[A] = \quad (63)$$

$$[G] = [\bar{G}] \quad (64)$$

For any given acceleration at the fluid–wall interface, Eq. (60) is solved to obtain the hydrodynamic pressure within the fluid domain. For the nonlinear analysis of the fluid domain to obtain hydrodynamic pressure, the steps to be followed are given in the form of flow chart in Fig. 2.

3. Theoretical formulation for wall

The equation of motion of a structure subjected to external forces can be written in standard finite element form as

$$[M_i]\{\ddot{u}_i\} + [C_i]\{\dot{u}_i\} + [K_i]\{u_i\} = \{F_{di}\} \quad (65)$$

where $i = l$ and r for left and right wall respectively and $\{\ddot{u}_i\}$, $\{\dot{u}_i\}$ and $\{u_i\}$ are nodal accelerations, velocities and displacements, $\{F_{di}\}$ is the nodal forces. Mass matrix, $[M_i]$ and stiffness $[K_i]$ may be expressed as follows.

$$[M_i] = \iiint_V [N_w]^T \rho [N_w] dv \quad \text{and} \quad [K_i] = \iiint_V [B^T][D][B] dv \quad (66)$$

where $[N_w]$ is shape function matrix, $[B]$ is strain displacement matrix, $[D]$ is strain displacement matrix and ρ is density of the wall. However, in present investigation, the tank walls have been discretised by two dimensional eight node rectangular elements and walls are assumed to behave linearly and are in a state of plane strain. The Rayleigh damping is considered as structural damping. Therefore, the damping matrix, $[C_i]$ can be expressed as

$$[C] = a'[M] + b'[K] \quad (67)$$

Here, a' and b' are called the proportional damping constants. The relationship between a' , b' and the fraction of critical damping at a frequency ω is given by the following equation.

$$\xi' = \frac{1}{2} \left(a'\omega + \frac{b'}{\omega} \right) \quad (68)$$

Damping constants a' and b' are determined by choosing the fraction of critical damping ξ'_1 and ξ'_2 at two different frequencies ω_1 & ω_2 and solving simultaneously equations a' and b' . Thus,

$$a' = \frac{2(\xi'_2\omega_2 - \xi'_1\omega_1)}{(\omega_2^2 - \omega_1^2)} \quad (69)$$

$$b' = \frac{2\omega_1\omega_2(\xi'_2\omega_1 - \xi'_1\omega_2)}{(\omega_2^2 - \omega_1^2)}$$

Usually, ω_1 is taken as the lowest natural frequency of the structure, and ω_2 is the highest frequency of interest in the loading or response. In the present study, the fraction of critical damping for both the frequencies are chosen as the same i.e. $\xi'_1 = \xi'_2 = \xi'$. Thus, above equation may be expressed as

$$\begin{aligned} a' &= \frac{2\xi'}{(\omega_2 + \omega_1)} \\ b' &= \frac{2\xi'\omega_1\omega_2}{(\omega_2 + \omega_1)} \end{aligned} \quad (70)$$

4. Coupling of fluid–structure system

In the fluid–structure interaction problems, the structure and the fluid do not vibrate as separate systems under external excitations, rather they act together in a coupled way. Therefore, the fluid–structure interaction problem has to be dealt in a coupled way. In the present study, a direct coupling approach is developed to get the coupled tank–water response under external excitation. The coupling of tank walls and water domain may be formulated in following way. The discrete structural equation for left and right walls may be written respectively as:

$$[M_l]\{\ddot{u}_l\} + [C_l]\{\dot{u}_l\} + [K_l]\{u_l\} - [Q_l]\{p\} - \{F_{dl}\} = 0 \quad (71)$$

$$[M_r]\{\ddot{u}_r\} + [C_r]\{\dot{u}_r\} + [K_r]\{u_r\} - [Q_r]\{p\} - \{F_{dr}\} = 0 \quad (72)$$

Here, subscripts l and r stand for left and right wall respectively. The coupling term $[Q]$ (Zienkiewicz and Newton [25]) arises to satisfy the compatibility condition at the tank wall–water interfaces. In Eqs. (71) and (72) the term $[Q_i]\{p\}$, where $i = l$ for left wall and $i = r$ for right wall comes to take care the addition force due to the hydrodynamic pressure within the water, adjacent to the tank walls. Similarly, in Eq. (74), the term $[Q_i^T]\{\ddot{u}_i\}$, is essential to take the effect of additional pressure due to the acceleration of walls, as the flexibility of the walls is considered in present analysis.

$$\int_{\Gamma_s} N_s^T n p d\Gamma = \left(\int_{\Gamma_s} N_s^T n N_f d\Gamma \right) p = [Q]\{p\} \quad (73)$$

where n is the direction vector of the normal to the tank wall–water interface. N_s and N_f are the shape functions of structure and fluid respectively. Similarly, the finite element equation for the fluid with elastic structure may be written using Eq. (60) as

$$[E]\{\ddot{P}\} + [A]\{\dot{P}\} + [G]\{P\} + [Q_l^T]\{\ddot{u}_l\} + [Q_r^T]\{\ddot{u}_r\} = \{F_l\} - \{F_{nl}\} \quad (74)$$

Now, the system of Eqs. (71), (72) and (74) are coupled in a second-order ordinary differential equations, which defines the coupled fluid–structure system completely. These equations may be written in a combined form as

$$\begin{aligned} &\begin{bmatrix} M_l & 0 & 0 \\ Q_l^T & E & Q_r^T \\ 0 & 0 & M_r \end{bmatrix} \begin{Bmatrix} \ddot{u}_l \\ \ddot{P} \\ \ddot{u}_r \end{Bmatrix} + \begin{bmatrix} C_l & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & C_r \end{bmatrix} \begin{Bmatrix} \dot{u}_l \\ \dot{P} \\ \dot{u}_r \end{Bmatrix} \\ &+ \begin{bmatrix} K_l & -Q_l & 0 \\ 0 & G & 0 \\ 0 & -Q_r & K_r \end{bmatrix} \begin{Bmatrix} u_l \\ p \\ u_r \end{Bmatrix} = \begin{Bmatrix} F_{dl} \\ F_l - F_{nl} \\ F_{dr} \end{Bmatrix} \end{aligned} \quad (75)$$

Neglecting nonlinear convective acceleration, Eq. (75) becomes

$$\begin{aligned} &\begin{bmatrix} M_l & 0 & 0 \\ Q_l^T & E & Q_r^T \\ 0 & 0 & M_r \end{bmatrix} \begin{Bmatrix} \ddot{u}_l \\ \ddot{P} \\ \ddot{u}_r \end{Bmatrix} + \begin{bmatrix} C_l & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & C_r \end{bmatrix} \begin{Bmatrix} \dot{u}_l \\ \dot{P} \\ \dot{u}_r \end{Bmatrix} \\ &+ \begin{bmatrix} K_l & -Q_l & 0 \\ 0 & G & 0 \\ 0 & -Q_r & K_r \end{bmatrix} \begin{Bmatrix} u_l \\ p \\ u_r \end{Bmatrix} = \begin{Bmatrix} F_{dl} \\ F_l \\ F_{dr} \end{Bmatrix} \end{aligned} \quad (76)$$

4.1. Computation of velocity and displacement of fluid

The accelerations of the fluid particles can be calculated after computing the hydrodynamic pressure in the water tank. Also, the velocity of the fluid particle may be calculated from the known values of acceleration at any instant of time using Gill's time integration scheme [26] which is a step-by-step integration procedure based on Runge–Kutta method [27]. This procedure is advantageous over other available methods as it (i) needs less storage registers, (ii) controls the growth of rounding errors, (iii) is usually stable and (iv) is computationally economical. At any instant of time t , velocity will be

$$V_t = V_{t-\Delta t} + \Delta t \dot{V}_t \quad (77)$$

Velocity vectors in the fluid domain may be plotted based on velocities computed at the Gauss points of each individual element and then extrapolating to their nodal points. Similarly, the displacement of fluid particles in the fluid domain, U at every time instant can also be computed as

$$U_t = U_{t-\Delta t} + \Delta t V_t \quad (78)$$

5. Numerical results

5.1. Selection of time step

In order to choose a time step for the solution of the tank–water system by the Newmark's integration technique, a tank–water system with following properties is considered. Tank wall: height (H_t) = 2 m, length (L) = 0.8 m, thickness of wall = 50 mm, mass density (ρ_t) = 7800 kg/m³, Poisson's ratio (ν) = 0.3, modulus of elasticity (E_t) = 200 GPa and structural damping = 5%; Water: depth (H_f) = 1.6 m, acoustic speed (C) = 1440 m/s, mass density (ρ_f) = 1000 kg/m³. The study is carried out for three different exciting frequencies, TC/H_f as 10, 100 and 1000 with amplitude of 5.0 g. Here the flexible walls and fluid are discretized by 1×10 (i.e., $N_h = 1$ and $N_v = 10$) and 4×8 (i.e., $N_h = 4$ and $N_v = 8$) respectively. The maximum pressure coefficient ($C_p = p/\rho_f aH_f$) for different number of time step (N_t) at different exciting frequencies are summarized in Table 1. It is observed from the tabular results that the developed hydrodynamic pressure for different values of TC/H_f is converged for values of $N_t = 32$. Thus, the time step (Δt) for the analysis of tank–water system is adopted as $T/32$ for all the case unless it is mentioned.

5.2. Nonlinear responses of tank–water system

In order to investigate the effects of nonlinear convective acceleration, few numerical examples related to water tank are studied. In the present analysis, the number of time steps per cycle of excitation for solution of tank–wall system by the Newmark's, N_t is considered as 32 at which the results are found to converge sufficiently as shown in Table 1. Thus the time step for the analysis of water tank is taken as $T/32$.

Table 1

Convergence of hydrodynamic pressure coefficients (C_p) for different time steps.

N_t	$TC/H_f = 10$	$TC/H_f = 100$	$TC/H_f = 1000$
8	0.358	2.674	2.241
16	0.357	3.302	2.368
24	0.356	3.356	2.390
32	0.356	3.361	2.395
64	0.356	3.361	2.395
128	0.356	3.361	2.395

Table 2

First three natural time periods of the fluid in the tank.

Mode number	Natural time period in sec	
	Present study	Virella et al. [22]
1	8.46	8.38
2	3.94	3.70
3	2.89	2.78

Example 1. Validation of the proposed algorithm.

In order to validate the proposed algorithm a bench marked problem is solved and results are compared with the results obtained by Virella et al. [22]. The geometric and material properties of the tank are considered as follows: height of water in the tank (H) = 6.10 m, length of tank (R) = 30.5 m, so that ratio of height to length (H/R) = 0.4, density of water = 983 kg/m³, pressure wave velocity = 1451 m/s. Here, the interaction of fluid-tank system is neglected and the fluid is discretized by 4×8 (i.e., $N_h = 4$ and $N_v = 8$). The first three natural time periods of the fluid in the tank are listed and compared with those values obtained by Virella et al. [21] in Table 2. The tabulated results show the accuracy of the present method.

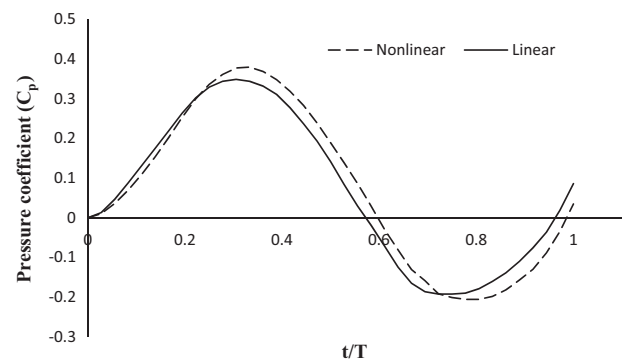
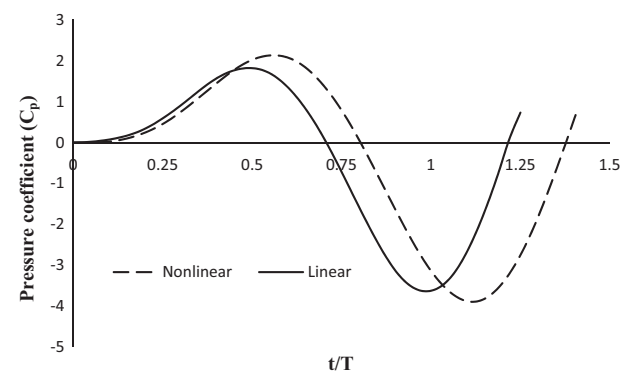
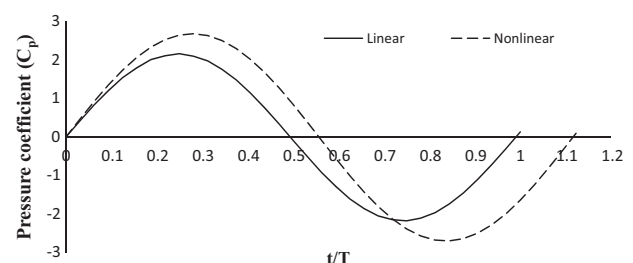
Example 2. Effect of excitation frequency on tank wall.

The geometry of water tank as shown in Fig. 1 is considered here. The dimensions and different properties of tank-fluid system are as follows: Tank wall: height (H_t) = 2 m, length (L) = 0.8 m, thickness of wall = 50 mm, mass density (ρ_t) = 7800 kg/m³, Poisson's ratio (ν) = 0.3, modulus of elasticity (E_t) = 200 GPa and structural damping = 5%; Water: depth (H_f) = 1.6 m, acoustic speed (C) = 1440 m/s, mass density (ρ_f) = 1000 kg/m³. The study is carried out for three different excitation frequencies, TC/H_f as 10, 100 and 1000, where T is the time period. Here, the flexible walls and fluid are discretized by 1×10 (i.e., $N_h = 1$ and $N_v = 10$) and 4×8 (i.e., $N_h = 4$ and $N_v = 8$) respectively. A sinusoidal acceleration of amplitude 5.0 g is applied on the tank walls. For nonlinear analysis, the tolerance value is considered as 10^{-8} to obtain high level of accuracy. Maximum hydrodynamic pressures considering linear and nonlinear analysis for different values of TC/H_f are listed in Table 3. The tabular results depict the increase of nonlinear pressure of 2.0%, 5.1% and 10.3% over the linear hydrodynamic pressure for TC/H_f as 10, 100 and 1000 respectively. The variations of linear and nonlinear hydrodynamic pressure at point A (Fig. 1) for different TC/H_f are shown in Figs. 3–5. From Fig. 3, it is observed that for $TC/H_f = 10$, the maximum nonlinear pressure at the time of 0.75T does not differ much with that of maximum linear pressure, even though the variations of linear and nonlinear pressure are quite different. On the other hand, a sharp deviation between linear and nonlinear pressure is noticed all through the time for $TC/H_f = 100$ and 1000 (Figs. 4 and 5) which signifies the importance of frequency of excitation in nonlinear analysis of flexible water tank. The velocity vectors in the water domain are plotted in Fig. 6. The vertical axis represents the height of the water and the horizontal axis represents the length between two vertical walls of the tank. Fig. 6(b) and (c) clearly show that the fluid generates a tendency to rotate more in case of nonlinear analysis and

this rotating tendency increases the nonlinear hydrodynamic pressure. However, for $TC/H_f = 10$, the velocity plot from nonlinear analysis is almost similar to that obtained from linear analysis (Fig. 6(a)) which keeps both the pressures same.

Example 3. Effect of amplitude of excitation on tank wall.

Geometric and material properties for this problem are considered same as in Example 2. Similar to Example 2, the flexible walls and fluid are discretized by 1×10 (i.e., $N_h = 1$ and $N_v = 10$) and 4×8 (i.e., $N_h = 4$ and $N_v = 8$) respectively. Sinusoidal acceleration of three different amplitudes: 1.0 g, 2.5 g and 5.0 g are applied on the tank wall. The study is carried out for three different frequencies, TC/H_f as 10, 100 and 1000 and the tolerance value is considered as 10^{-8} . Linear and nonlinear hydrodynamic pressures along the face of tank wall for different TC/H_f are shown in Figs. 7–9. The results show that the hydrodynamic pressure along the face of the wall is increased when the nonlinearity effect of water is considered. From these figure, it is also noticed that nonlinear hydrodynamic pressure is almost equal to the linear hydrodynamic

**Fig. 3.** Hydrodynamic pressure at Point A for $TC/H_f = 10$.**Fig. 4.** Hydrodynamic pressure at Point A for $TC/H_f = 100$.**Fig. 5.** Hydrodynamic pressure at Point A for $TC/H_f = 1000$.**Table 3**Linear and nonlinear pressure for different TC/H_f .

TC/H_f	Nonlinear pressure coefficient ($P/(a\rho H_f)$)	Linear pressure coefficient ($P/(a\rho H_f)$)	% of increases in pressure coefficient
10	0.3561	0.3491	2.0
100	3.361	3.190	5.1
1000	2.395	2.172	10.3

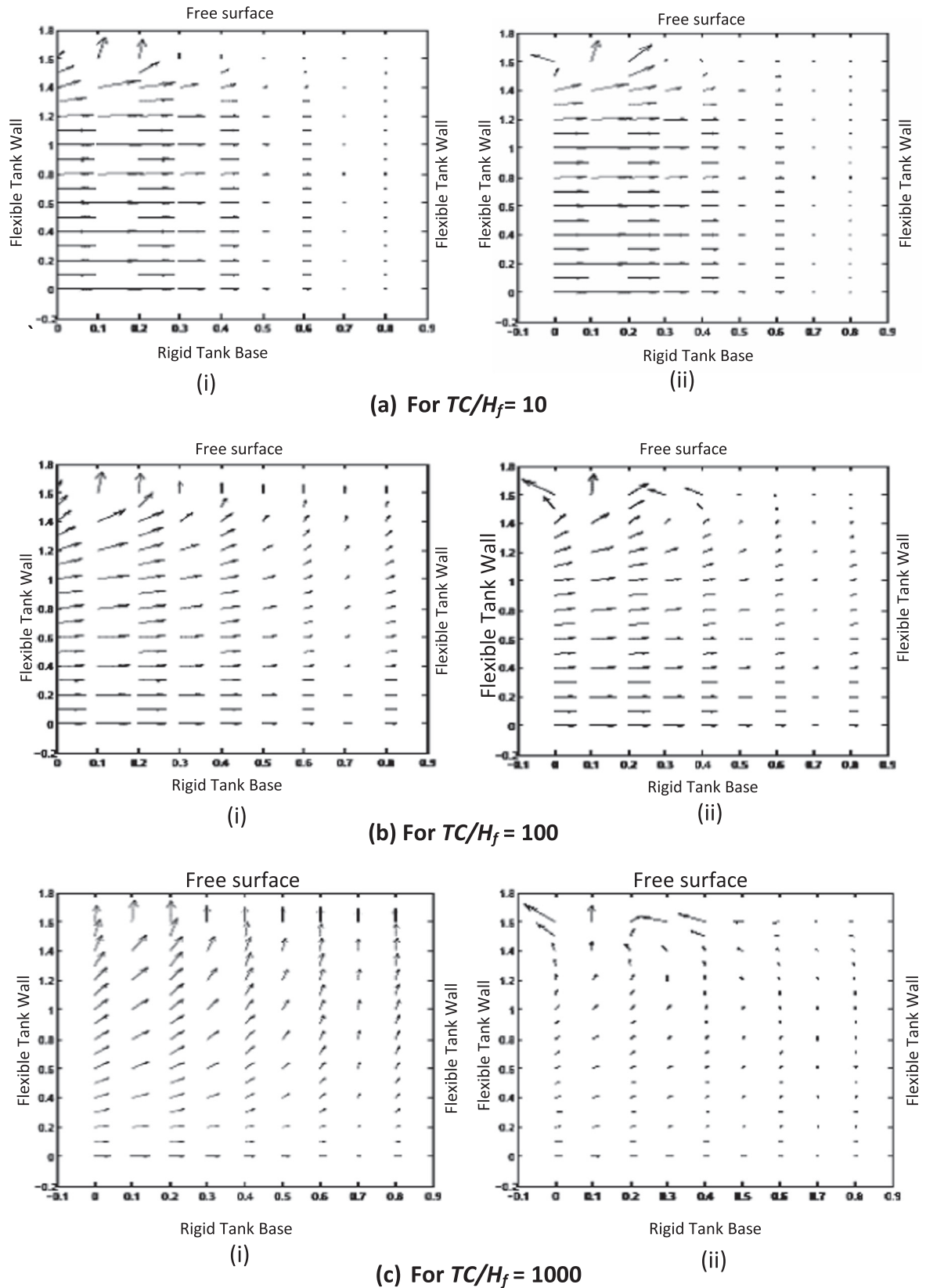


Fig. 6. Velocity vectors in fluid form (i) linear and (ii) nonlinear analysis.

pressure along the face of the tank for amplitude of excitation equal to 1.0 g. However, for other two amplitudes of excitation, i.e. for 2.5 g and 5 g a remarkable increase in nonlinear

hydrodynamic pressures are observed and the maximum increase is noticed for amplitude of 5.0 g. Sloshing in terms of vertical displacement coefficient (d/a), is indicated in Figs. 10–12 for

$TC/H_f = 10, 100$ and 1000 respectively. The sloshing for $TC/H_f = 1000$ is increased with the increases of amplitude of excitation which enhances the nonlinear hydrodynamic pressure (Fig. 12). Whereas, for $TC/H_f = 10$ and 100 sloshing remains almost same for all values of amplitude of excitation (Figs. 10 and 11), as a result no changes in nonlinear hydrodynamic pressure for different amplitudes of excitation.

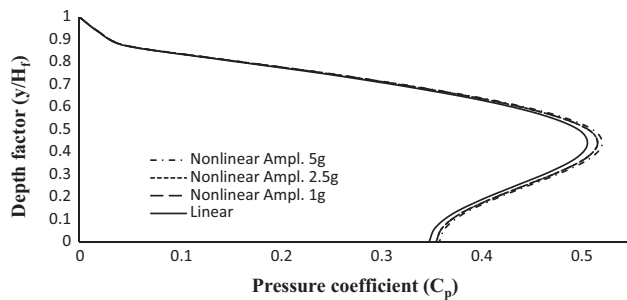


Fig. 7. Hydrodynamic pressure along the face of the wall for $TC/H_f = 10$.

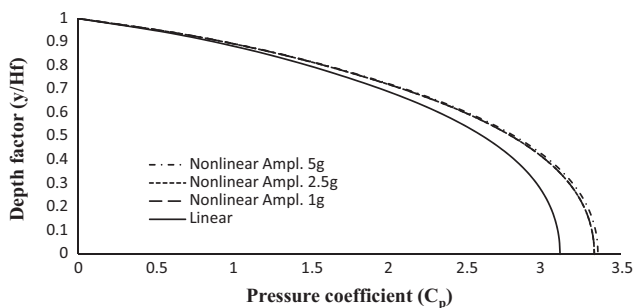


Fig. 8. Hydrodynamic pressure along the face of the wall for $TC/H_f = 100$.

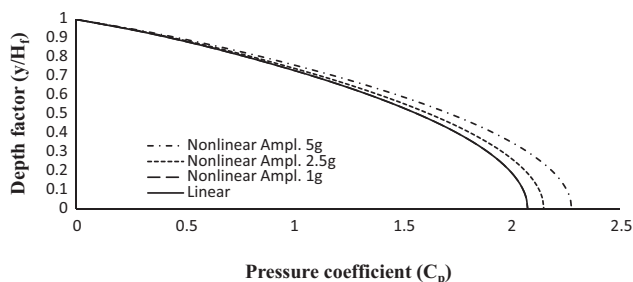


Fig. 9. Hydrodynamic pressure along the face of the wall for $TC/H_f = 1000$.

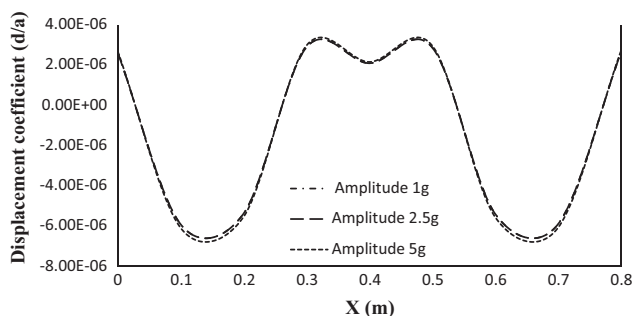


Fig. 10. Sloshed displacement for different amplitude of excitation for $TC/H_f = 10$.

Example 4. Effect of nonlinearity for different tank lengths.

Geometric and material properties considered in this case are same as considered in Example 2 except the length between two flexible walls of tank. Nonlinear behavior of water is studied for three different tank lengths, i.e. 0.8 m, 1.6 m, and 2.4 m and discretized by 4×8 (i.e., $N_h = 4$ and $N_v = 8$), 8×8 (i.e., $N_h = 8$ and $N_v = 8$) and 9×6 (i.e., $N_h = 12$ and $N_v = 8$) respectively. Two flexible walls of the tank are discretized by 1×10 (i.e., $N_h = 1$ and $N_v = 10$). The study is carried out for three different frequencies, TC/H_f as $10, 100$ and 1000 with a sinusoidal excitation of amplitude 5.0 g. Tolerance value for nonlinear hydrodynamic pressure is considered same as earlier. Linear and nonlinear hydrodynamic pressures at Point A (Fig. 1) for different cases are listed in Table 4. The tabular results depict that nonlinear behavior of fluid depends on frequency i.e. TC/H_f of excitation which supports the finding of previous examples. For $TC/H_f = 10$ and 100 nonlinear effect of fluid is not significant for tank length of $1H_f$ and $1.5H_f$. However, for tank length of $0.5H_f$, a considerable increase in nonlinear pressure is

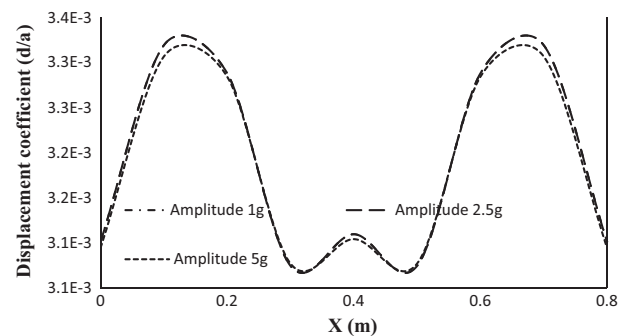


Fig. 11. Sloshed displacement for different amplitude of excitation for $TC/H_f = 100$.

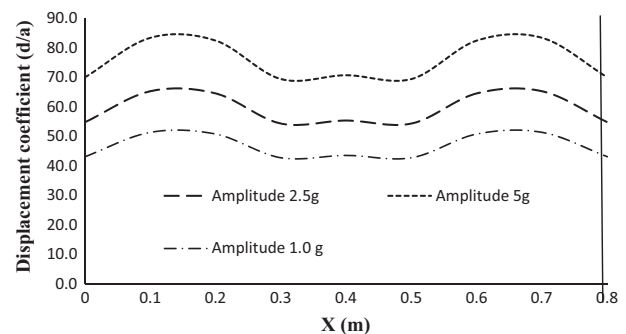


Fig. 12. Sloshed displacement for different amplitude of excitation for $TC/H_f = 1000$.

Table 4

Linear and nonlinear pressure for different reservoir length.

TC/H_f	Tank length	Nonlinear hydrodynamic pressure coefficient ($P/(\rho g H_f)$)	Linear hydrodynamic pressure coefficient ($P/(\rho g H_f)$)	% of increase in pressure coefficient
10	$0.5H_f$	0.356	0.349	2.0
	$1H_f$	0.256	0.255	0.4
	$1.5H_f$	0.241	0.241	0.0
100	$0.5H_f$	3.361	3.190	5.1
	$1H_f$	1.981	1.958	1.2
	$1.5H_f$	1.477	1.465	0.8
1000	$0.5H_f$	2.395	2.172	10.3
	$1H_f$	1.226	1.152	6.4
	$1.5H_f$	0.946	0.902	4.9

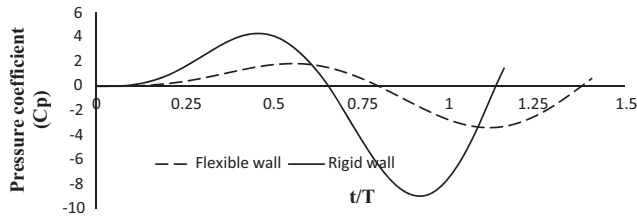


Fig. 13. Comparisons of hydrodynamic pressure for tank length of $0.5H_f$ and $TC/H_f = 100$.

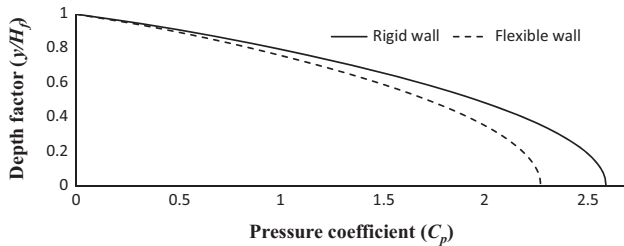


Fig. 14. Comparisons of hydrodynamic pressure along the face of wall for tank length of $0.5H_f$ and $TC/H_f = 1000$.

observed. At $TC/H_f = 1000$, nonlinear hydrodynamic pressure is increased for all lengths of the water tank. However, the % increase in nonlinear pressure over linear pressure is decreased as the length of the tank increases. Hydrodynamic pressures are also compared with those for rigid wall (Figs. 13 and 14). In this case, the tank length is considered as 0.8 m. Hydrodynamic pressure at point A for rigid tank walls is almost three times to those of flexible wall case (Fig. 13). The reflection of pressure wave from rigid walls

is comparatively more; as a result the hydrodynamic pressure increases. Similar trend for hydrodynamic pressure along the face of the wall is noticed in Fig. 14.

Example 5. Behavior of water tank under earthquake excitation.

Nonlinear behavior of water tank under earthquake excitation is studied in this section. Two different tanks of length 0.8 m and 1.6 m are considered. Tank height and depth of water with in the tank are considered as 2 m and 1.6 m respectively. For finite element implementation fluid domain are discretized by 4×8 (i.e., $N_h = 4$ and $N_v = 8$), 8×8 (i.e., $N_h = 8$ and $N_v = 8$) respectively. Similarly, elastic tank walls are discretized by 1×10 (i.e., $N_h = 1$ and $N_v = 10$). Fluid and tank wall properties in this case are same as considered in Example 2. Koyna earthquake acceleration is considered as external excitation. Comparison between linear and nonlinear hydrodynamic pressure at Point A (Fig. 1) of the tank is made in Figs. 15 and 16 for tank of length of 0.8 m and 1.6 m respectively. From these graphical results, it is noted that increase in hydrodynamic pressure due to nonlinearity of fluid is insignificant for the both lengths of the tank.

6. Conclusions

This paper presents a nonlinear hydrodynamic analysis of elastic water tank of different lengths against horizontal harmonic excitation of various frequencies and amplitudes. The numerical exercises show that the amplitude, frequency and the length of the tank may alter the responses of the water domain. The outcome of the analysis results show that the nonlinear convective term should not be neglected when excitation frequencies are lower than the first fundamental frequency of the water domain. Also, the nonlinear effect should not be ignored for relatively narrow water tank. The differences between linear and nonlinear hydrodynamic pressure become more if the magnitude of the

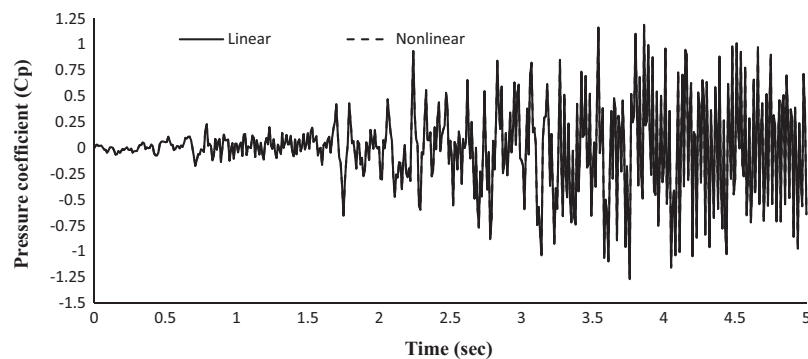


Fig. 15. Hydrodynamic pressure due to Koyna earthquake for tank of length $0.5H_f$.

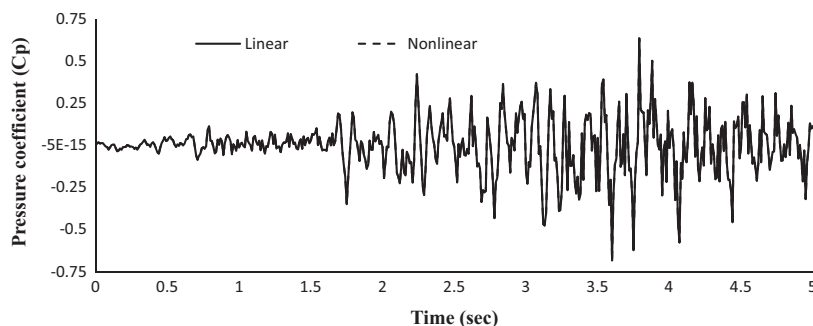


Fig. 16. Hydrodynamic pressure against Koyna earthquake for tank of length H_f .

acceleration is higher. However, the linear and nonlinear hydrodynamic pressures are almost of same magnitude in case of earthquake excitation.

References

- [1] Housner GW. The dynamic behavior of water tanks. *Bull Seismol Soc Am* 1963;53(2):381–9.
- [2] Fenves G, Vargas LM-Loll. Nonlinear dynamic analysis of fluid–structure systems. *J Eng Mech, Am Soc Civ Eng* 1988;114:219–40.
- [3] Zienkiewicz OC, Paul DK, Hinton E. Cavitation in fluid–structure response (with particular reference to dams under earthquake loading). *Earthquake Eng Struct Dynam* 1983;11:463–81.
- [4] Pelecanos N, Kontoe S, Zdravkovic L. Numerical modeling of hydrodynamic pressures on dams. *J Comp Geotech* 2013;53:68–82.
- [5] Bouaanani N, Lu FY. Assessment of potential-based fluid finite elements for seismic analysis of dam–reservoir systems. *J Comp Struct* 2009;87:206–24.
- [6] Maity D, Bhattacharyya SK. A parametric study on fluid–structure interaction problems. *J Sound Vib* 2003;263:917–35.
- [7] Gogoi I, Maity D. A novel procedure for determination of hydrodynamic pressure along upstream face of dams due to earthquakes. *J Comp Struct* 2010;88:539–48.
- [8] Tung CC. Hydrodynamic forces on submerged vertical circular cylindrical tanks under ground excitation. *Appl Ocean Res* 1979;1:75–8.
- [9] Williams AN, Moubayed WI. Earthquake-induced hydrodynamic pressures on submerged cylindrical storage tanks. *J Ocean Eng* 1990;17(3):181–99.
- [10] Choun YS, Yun CB. Sloshing characteristic in rectangular tanks with a submerged block. *J Comp Struct* 1996;61(3):401–13.
- [11] Panigrahy PK, Saha UK, Maity D. Experimental studies on sloshing behavior due to horizontal movement of liquids in baffled tanks. *J Ocean Eng* 2009;36:213–22.
- [12] Hua CW, Fuh CB, Kan HT. Hydrodynamic forces induced by transient sloshing in a 3D rectangular tank due to oblique horizontal excitation. *J Comp Math Appl* 2013;65:1163–86.
- [13] Haroun MA, Tayel MA. Response of tank to vertical seismic excitation. *J Earthquake Eng Struct Dyn* 1985;13:583–95.
- [14] Veletsos AS, Tang Y. Dynamics of vertically excited liquid storage tanks. *J Struct Eng, ASCE* 1986;112(6):1228–46.
- [15] Haroun MA, Abou WI. Parametric study of seismic soil–tank interaction II: Vertical excitation. *J Struct Eng, ASCE* 1992;118(3):798–812.
- [16] Kianoush MR, Chen JZ. Effect of vertical acceleration on response of concrete rectangular liquid storage tanks. *J Eng Struct* 2006;28:704–15.
- [17] Park JH, Koh HM, Kim J. Liquid–structure interaction analysis by coupled boundary element–finite element method in time domain. *Proceedings of 7th international conference on boundary element technology*. Southampton (England): Computational Mechanics Publication; 1990. p. 89–92.
- [18] Chen JZ, Kianoush MR. Generalized SDOF system for seismic analysis of concrete rectangular liquid storage tanks. *J Eng Struct* 2009;31:2426–35.
- [19] Moslemi M, Kianoush MR. Parametric study on dynamic behavior of cylindrical ground-supported tanks. *J Eng Struct* 2012;42:214–30.
- [20] Celebi MS, Akyildiz H. Nonlinear modeling of liquid sloshing in a moving rectangular tank. *J Ocean Eng* 2002;29(12):1527–53.
- [21] Barrios HH, Zavoni EH, Alvaro A, Rodriguez A. Nonlinear sloshing response of cylindrical tanks subjected to earthquake ground motion. *J Eng Struct* 2007;29:3364–76.
- [22] Virella JC, Prato CA, Godoy LA. Linear and nonlinear 2D finite element analysis of sloshing modes and pressures in rectangular tanks subject to horizontal harmonic motions. *J Sound Vib* 2008;312:442–60.
- [23] Lai WM, Rubin D, Krempf E. *Introduction to continuum mechanics*. 4th ed. Butterworth-Heinemann, Elsevier; 2010.
- [24] Chen BF. Nonlinear hydrodynamic effects on concrete dam. *J Eng Struct* 1996;18(3):201–12.
- [25] Zienkiewicz OC, Newton RE. Coupled vibration of a structure submerged in a compressible fluid. In: *Proceedings of international symposium on finite element techniques*. Stuttgart; 1969.
- [26] Gill S. A process for the step-by-step integration of differential equations in an automatic digital computing machine. *Proc Cambridge Philos Soc* 1951;47:96–108.
- [27] Ralston A, Wilf HS. *Mathematical models for digital computers*. New York: Wiley; 1965.